

Econ 381: Introduction to Econometrics

Handout 4: Review of Statistics

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Regression estimates change from sample to sample

In Handout 2, we learned that the OLS slope

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}$$

is computed from a sample. A different random sample generally produces a different slope.

How do we quantify that randomness?

- ▶ Standard errors
- ▶ Confidence intervals
- ▶ Hypothesis tests

Statistics connects the estimate we observe to the uncertainty created by sampling.

Our roadmap: from population quantities to sample evidence

Before analyzing uncertainty in econometric problems, we review nine ideas:

1. A **population** and its expectation
2. A **random sample** from that population
3. The **sample average** as an estimator
4. The **Law of Large Numbers** as the reason the estimator works
5. **Conditional expectations** for subpopulations
6. **Variance and standard deviation** for dispersion
7. **Covariance and correlation** for co-movement
8. The **Central Limit Theorem** for sampling distributions
9. **Standard errors and confidence intervals** for uncertainty

We will use the textbook dataset `caschool.dta`: 420 California school districts, with district test scores, student–teacher ratios, and other characteristics.

The population is determined by the research question

Research question	Population and variable of interest
How is average test performance related to class size?	All California school districts; Y = district test score and X = student–teacher ratio.
How are workers' wages related to education?	All U.S. workers in the target period; Y = wage and X = years of education.

The same word “average” can describe different quantities. We must define *whose* outcomes and *which* time period belong to the population.

A population can be finite or extremely large

Finite population

All California school districts in a specified year form a finite collection of units.

Each district has variables such as its test score and student–teacher ratio.

Very large population

All U.S. workers in a target period can be so numerous that we often think of the population as effectively **infinite**.

Each worker has variables such as wage and education.

In either case, the population contains every unit relevant to the research question—not only the units that happen to enter our dataset.

Expectation is the population average of a variable

Choose a variable Y for every unit in the population. Its expectation is the average across *all* those units:

$$\mu_Y = E[Y].$$

For a finite population of N units,

$$E[Y] = \frac{1}{N} \sum_{i=1}^N Y_i.$$

Thus, $E[\text{TestScore}]$ averages scores across all relevant districts, while $E[\text{Wage}]$ averages wages across all relevant workers.

Expectations are fixed population quantities, but they are typically **unknown** to researchers. Why?

Why population expectations are usually unknown

Because a census of the target population is often infeasible:

- ▶ collection cost and time;
- ▶ nonresponse or inaccessible units;
- ▶ populations that change while data are collected.

Instead, researchers collect a **random sample**: each unit's inclusion is governed by a random sampling mechanism.

Before sampling, the researcher does not know which particular units will appear. That uncertainty makes sample statistics random variables.

A random sample includes some districts and leaves out others

Suppose researchers have money and staff to collect data from only 10 school districts. They randomly select 10 districts from the population.

5 districts appearing in a sample of 10

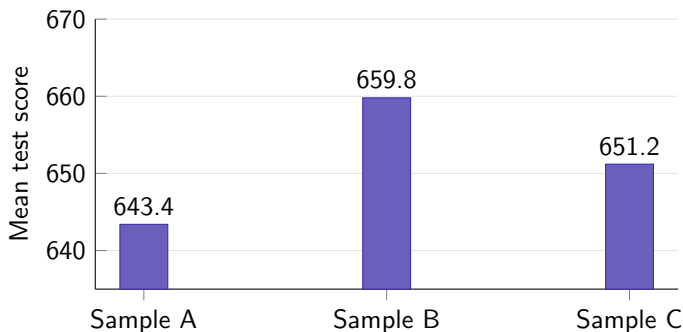
ID	County	District	Score
317	Sonoma	Mark West Union	666.85
190	Lassen	Shaffer Union	652.10
216	Fresno	Alvina	655.05
214	San Diego	Lakeside Union	654.85
270	Stanislaus	Gratton	661.90

5 districts not in it

ID	County	Score
227	San Bernardino	655.85
239	Humboldt	657.50
75	Santa Barbara	634.90
52	Madera	631.40
56	Los Angeles	632.00

Before drawing it, the researcher does not know which districts will be included.

Different random samples produce different averages



The three hypothetical samples have the **same sample size** and come from the **same population**, yet their sample averages differ. The researcher knows the sampling rule, not the realized sample in advance.

The sample average estimates the population expectation

Given a random sample $Y_1, \dots, Y_n$ from the population ($n < N$), the sample average is

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i.$$

Parameter

$\mu_Y = E[Y]$ is fixed but unknown.

Estimator

$\bar{Y}$ is observable but random across different samples of $\{Y_i\}_{i=1}^n$

For California schools, $\overline{\text{TestScore}}$ uses the sampled districts to estimate $E[\text{TestScore}]$.

A classroom benchmark: treat the 420 districts as the population

For this demonstration only, the 420 districts in `caschool.dta` form a finite teaching population. Its mean is visible to us:

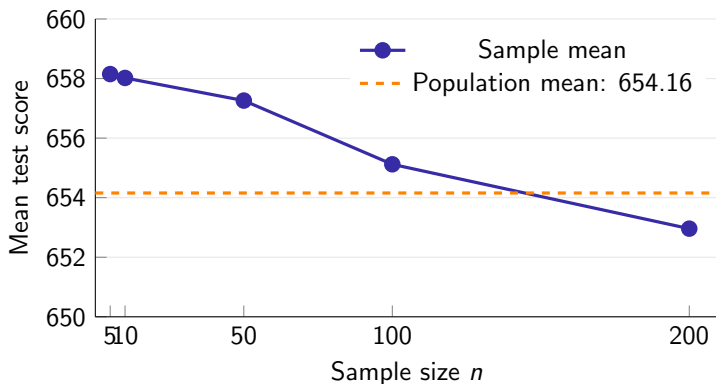
$$\mu_{\text{TestScore}} = 654.16.$$

We randomly order the districts once, then compute nested sample means using the first 5, 10, 50, 100, and 200 observations.

Why this device helps

In real research, the population mean is unknown. Here, knowing the benchmark lets us see how sampling behaves.

Larger samples move the sample mean toward the benchmark



Remark: The path need not approach the benchmark monotonically. Convergence is a probabilistic statement, not a promise that every larger sample is closer.

The Law of Large Numbers explains why averages stabilize

Technical statement: If $Y_1, \dots, Y_n$ are independent and identically distributed with a finite mean, then

$$\bar{Y} \xrightarrow{P} E[Y] \quad \text{as } n \rightarrow \infty.$$

Equivalently, for every $\varepsilon > 0$,

$$\text{Prob}(|\bar{Y} - E[Y]| > \varepsilon) \rightarrow 0.$$

Plain-language interpretation

As the random sample grows, sample average becomes increasingly representative for the population mean, so large discrepancies become less likely.

What the LLN does—and does not—tell us

It tells us

- ▶ why sample averages are useful;
- ▶ why estimates stabilize with more data;
- ▶ that large errors become less likely.

It does not tell us

- ▶ the exact error in our realized sample;
- ▶ that bias from nonrandom sampling disappears;
- ▶ how quickly uncertainty shrinks.

So we need tools that quantify sampling uncertainty: standard errors and confidence intervals (later in this handout), followed by hypothesis tests.

An overall average can hide important subgroup differences

Suppose we ask: *What is the average test score among districts in a particular county?*

County	Districts	Mean test score
Kern	27	639.29
Los Angeles	27	645.54
San Diego	21	660.42
Sonoma	29	664.54
Santa Clara	20	665.26
All observed districts	420	654.16

The relevant average depends on the information we condition on.

Conditional expectation averages within a subpopulation

Let C denote county. The conditional expectation

$$E[\text{TestScore} \mid C = c]$$

is the population average test score among districts in county c .

More generally, $E[Y \mid X = x]$ is the average of Y within the subpopulation satisfying $X = x$.

Unconditional versus conditional

$E[Y]$ averages across everyone in the population; $E[Y \mid X = x]$ holds X fixed and averages only within the specified group.

We can also condition on a continuous variable

The conditioning variable X need not define named groups such as counties. It may be continuous.

For example, let STR denote the student–teacher ratio. Then

$$E[\text{TestScore} \mid STR = x]$$

is the average test score among districts whose student–teacher ratio equals x .

Why this matters for econometrics

As x changes, the conditional expectation describes how the average outcome changes with an explanatory variable.

The regression function is a model for this relationship.

A subgroup sample average estimates a conditional expectation

In our sample/data, if n_c districts belong to county c , then

$$\bar{Y}_c = \frac{1}{n_c} \sum_{i: C_i = c} Y_i$$

estimates $E[Y \mid C = c]$.

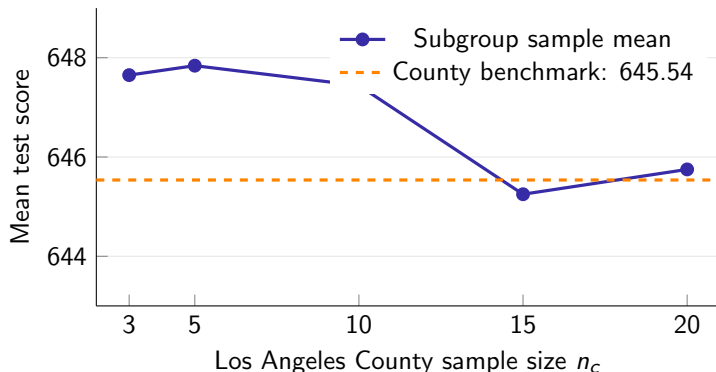
This is simply the sample average of Y computed using only observations with $C_i = c$; observations from other counties are left out of the sum.

For example, the sample mean among sampled Los Angeles County districts estimates

$$E[\text{TestScore} \mid C = \text{Los Angeles}].$$

The **estimator** $\bar{Y}_c$ remains **random**: a different sample usually contains different districts and produces a different subgroup average.

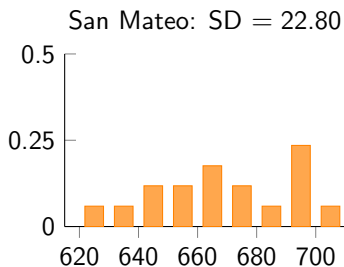
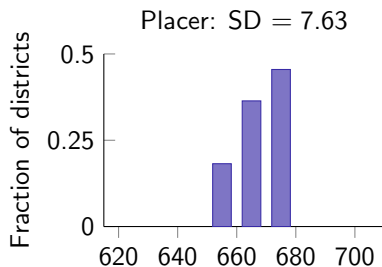
The LLN also works for conditional expectation



With a sufficiently large random sample from the county subpopulation, $\bar{Y}_c$ converges to $E[Y \mid C = c]$.

Averages do not describe how spread out outcomes are

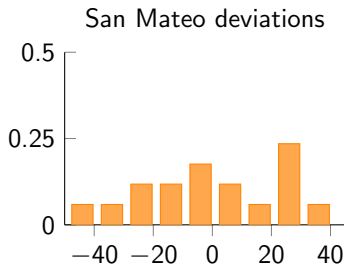
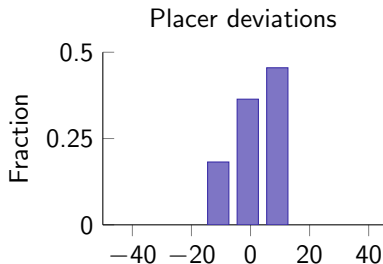
Two groups can have similar means but very different dispersion. In `caschool.dta`, district test scores are much more dispersed in San Mateo County than in Placer County.



Both histograms use identical axes and 10-point bins. Placer is concentrated near its mean; San Mateo spans almost the entire displayed range.

Dispersion is the spread of deviations around zero

For each population unit, center the outcome by subtracting its population mean: $Y - E[Y]$.



Placer's deviations concentrate near zero. San Mateo's deviations are widely dispersed on both sides of zero, so its **variance** is larger.

Variance averages squared deviations from the mean

Squaring $Y - E[Y]$ prevents negative and positive deviations from canceling and gives more weight to observations far from the mean.

The population **variance** of Y is

$$\text{Var}(Y) = E[(Y - E[Y])^2].$$

The population **standard deviation** is

$$\text{SD}(Y) = \sqrt{\text{Var}(Y)}.$$

Side note: Variance is in squared units; standard deviation returns to the original units of Y by doing the square root.

Sample variance applies the same idea to observed data

The sample variance and standard deviation are

$$s_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2, \quad s_Y = \sqrt{s_Y^2}.$$

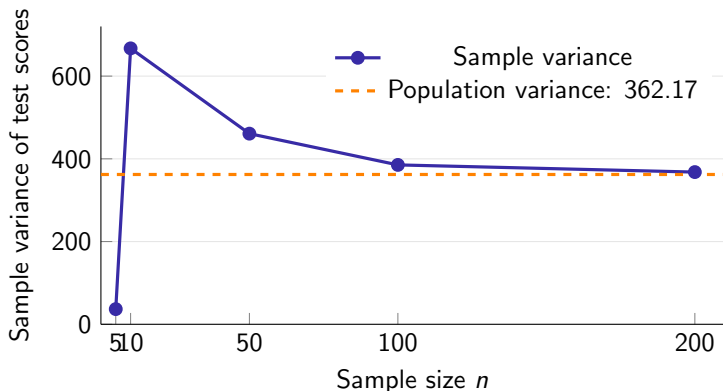
If sampled observations cluster near $\bar{Y}$, the squared deviations and s_Y^2 are small. If they are widely spread around $\bar{Y}$, s_Y^2 is large.

Technical note: Why $n - 1$? Estimating $\bar{Y}$ uses one degree of freedom; the correction makes s_Y^2 unbiased under random sampling.

California test scores

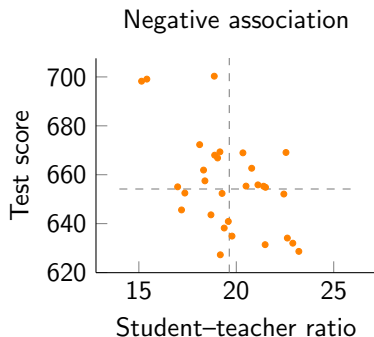
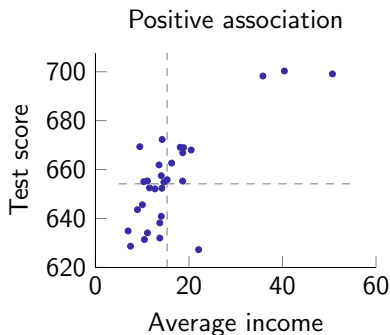
Across the 420 observed districts, the sample variance (divisor 419) is 363.03 and the sample SD is 19.05 points.

Sample variance also stabilizes as the sample grows



Because variance is itself an average of squared deviations, the LLN also applies: $s_Y^2 \xrightarrow{P} \text{Var}(Y)$.

Covariance asks whether two variables move together



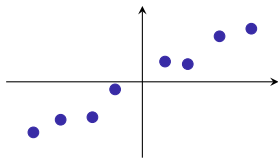
These plots show 30 randomly selected districts for demonstration; the numerical measures below use all 420.

Covariance records whether deviations have the same sign

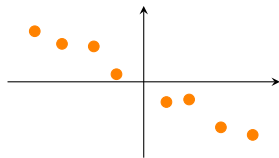
The population **covariance** is

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])].$$

Same signs: $\text{Cov} > 0$



Opposite signs: $\text{Cov} < 0$



Covariance averages the products of deviations. The CA data show the first pattern for income and test scores and the second for the student-teacher ratio and test scores.

Covariance changes with measurement units

Covariance inherits the units of both variables. If X is rescaled by a constant a , then

$$\text{Cov}(aX, Y) = a \times \text{Cov}(X, Y).$$

This property can be annoying when using covariance.

For example, considering test score and average income,

- ▶ Income measured in thousands of dollars: covariance = 97.85.
- ▶ Income measured in dollars: covariance = 97,852.
- ▶ The number changes by a factor of 1,000 even though the underlying relationship is identical.

Therefore, covariance is useful for determining the **direction** of co-movement, but its numerical *magnitude cannot be compared* across variables measured in different units.

Correlation is unit-free

Correlation divides by both standard deviations and removes measurement units:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)}, \quad -1 \leq \text{Corr}(X, Y) \leq 1.$$

- ▶ Correlation = +1: all observations lie on a straight line with positive slope.
- ▶ Correlation = -1: all observations lie on a straight line with negative slope.
- ▶ 0: no linear co-movement. Values closer to ± 1 indicate stronger linear association.

The income–test score correlation remains 0.712 whether income is measured in dollars or thousands of dollars.

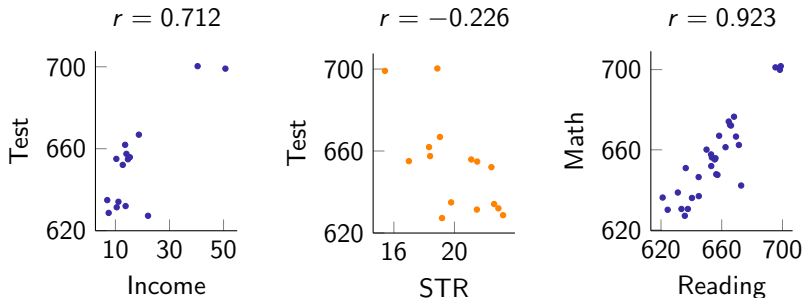
California districts show both signs

Variable pair	Covariance	Correlation
Test score and average income	97.85	0.712
Test score and student-teacher ratio	-8.14	-0.226
Reading and mathematics scores	347.21	0.923

The sign describes direction. The magnitude of covariance depends on measurement units, while correlation is unit-free and therefore easier to compare across pairs.

Remark: Like regressions, neither measure by itself establishes causality.

The three variable pairs show different linear patterns



Reading and mathematics scores cluster most tightly around an upward-sloping line

Test scores and the student-teacher ratio show the weakest linear pattern.

Sample covariance and correlation estimate population co-movement

The sample covariance is

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}),$$

and the sample correlation is

$$r_{XY} = \frac{s_{XY}}{s_X s_Y}.$$

For test scores and average income, one nested random sample gives

n	5	10	50	100	200
r_{XY}	.758	.815	.758	.727	.702

The full-data benchmark is 0.712. The LLN also applies to the sample covariance and correlation: s_{XY} and r_{XY} converge to their population counterparts as sample size becomes larger

Review: concepts and estimators

Concept	Population quantity	Sample estimator
Overall average	$E[Y]$	$\bar{Y}$
Average when $X = x$	$E[Y X = x]$	subgroup average $\bar{Y}_x$
Dispersion	$\text{Var}(Y)$, $\text{SD}(Y)$	s_Y^2 , s_Y
Co-movement	$\text{Cov}(X, Y)$	s_{XY}
Linear association	$\text{Corr}(X, Y)$	r_{XY}

The Law of Large Numbers explains why these sample moments approach their population counterparts as the random sample grows.

Optional technical reference—not required

Stock and Watson, *Introduction to Econometrics*, Chapter 2:

- ▶ S2.2 (expectation and variance),
- ▶ S2.3 (conditional distributions, covariance, and correlation),
- ▶ S2.5 (random sampling and the sample average), and
- ▶ S2.6 (LLN and consistency).

The sample average is itself a random variable

The LLN tells us that $\bar{Y}$ approaches μ_Y . But different random samples give different averages: $\bar{Y}$ is also a random variable.

Imagine an unlimited number of student researchers. Each independently draws n observations from the same population and reports one average. Their averages form the sampling distribution of $\bar{Y}$.

Its center and spread

For independent, identically distributed (i.i.d.) observations with mean μ_Y and finite variance $\text{Var}(Y)$,

$$E[\bar{Y}] = \mu_Y, \quad \text{Var}(\bar{Y}) = \frac{\text{Var}(Y)}{n}, \quad \text{SD}(\bar{Y}) = \frac{\text{SD}(Y)}{\sqrt{n}}.$$

Larger samples make the students' estimates less spread out around the true mean. An unusually high or low observation (e.g., outliers) has less influence on an average when it is only one of many observations.

A university sends students to study California schools

Let us explore the sampling distribution with a thought experiment.

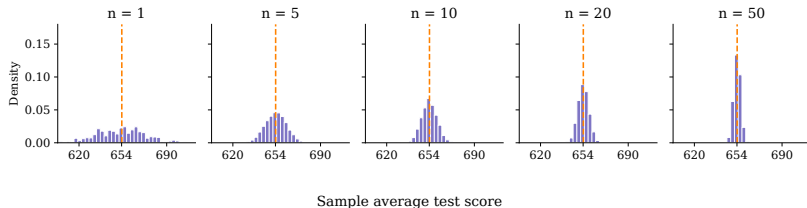
A university wants to learn the average test score and student–teacher ratio of our 420 California districts.

1. It can fund each student to collect n district observations.
2. Put the 420 district names in a bag. Each student draws a name, records that district's data, puts the name back, and mixes the bag before drawing again.
3. After n draws, each student calculates the two sample averages. The university collects results from 3,000 students.

Every district has the same chance on each draw. Putting names back keeps the draws independent; However, the same district may appear more than once.

For this data collection experiment, we know the full population's mean and SD, so we can check the students' estimates.

A larger survey makes students' averages less spread out



Each panel shows the **test-score averages** reported by 3,000 students with the same survey budget n . The dashed line is the true mean for all 420 districts, $\mu_Y = 654.16$.

With only one district, a student's answer depends heavily on which district is drawn. With more districts, high and low scores tend to balance out.

The center stays near μ_Y , while the spread shrinks as $SD(Y)/\sqrt{n}$.
Which survey budget would you choose if you wanted a more reliable estimate?

Put each student's estimation error on a common scale

A test-score error of 5 points and a student-teacher-ratio error of 5 are not equally large. We need a common scale to compare them.

$$Z_n = \frac{\bar{Y} - \mu_Y}{SD(Y)/\sqrt{n}}.$$

- ▶ Subtract μ_Y to measure the estimation error $\bar{Y} - \mu_Y$.
- ▶ Divide by the sampling SD of $\bar{Y}$, $SD(Y)/\sqrt{n}$, to express that error in units of typical sampling variation.
- ▶ $Z_n = 2$ means the estimate is two sampling SDs above the true mean; $Z_n = -2$ means two below it.

The university collects one Z-score from each of its 3,000 students for each variable. **What shape should these Z-scores have?**

Each variable uses its own population mean and SD. For test scores, these are approximately 654.16 and 19.03.

The CLT gives different averages the same normal limit

Central Limit Theorem (CLT)

For i.i.d. observations with finite, positive variance,

$$\frac{\sqrt{n}(\bar{Y} - \mu_Y)}{\text{SD}(Y)} \xrightarrow{d} N(0, 1) \quad \text{as } n \rightarrow \infty.$$

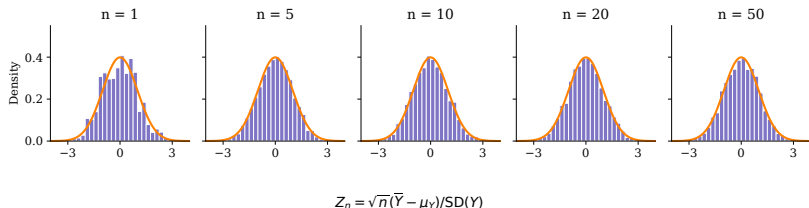
In words: as each student's sample grows, the distribution of the **standardized averages** approaches a standard normal bell curve.

The power of this result is that the original variable can have many different distributions: **discrete or continuous**, symmetric or skewed. Under these assumptions, its standardized average has the same limit.

For example, we could average a count, a test score, or a student–teacher ratio. The original data *need not* be normally distributed.

The result concerns **standardized sample averages**. Simply centering and rescaling individual observations does not make them normal.

Test-score averages become approximately normal



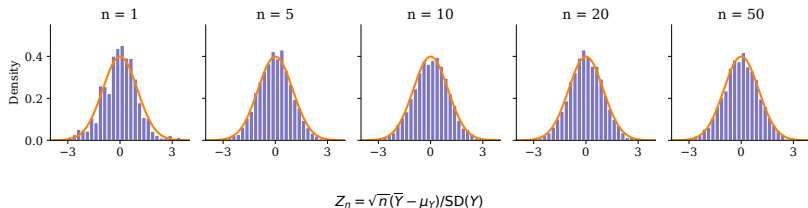
Blue: Z-scores from 3,000 students at each budget n . **Orange:** the same $N(0, 1)$ bell curve in every panel.

Most students draw a mix of high and low scores; very high or very low averages are less common.

The bell curve describes that pattern. Test scores already look fairly bell-shaped at $n = 50$, so the change here is modest.

Unlike the previous plot of raw averages, these plots do not keep getting narrower: we have divided by the shrinking sampling SD.

Student-teacher ratios have the same normal limit



Now look at the same 3,000-student thought experiment for student-teacher ratios, again with $n = 1, 5, 10, 20, 50$ districts per student.

The units and original distribution differ from test scores. But after centering and scaling their averages, the students' Z-scores approach the same $N(0, 1)$ curve.

For these district data, normality is an **approximation at every finite sample size**, even a large one. The CLT describes the limiting shape; it does not promise an exact normal distribution at any particular n .

The normal distribution: a formal definition

A continuous random variable W has a normal distribution with mean μ and variance $\sigma^2 > 0$, written $W \sim N(\mu, \sigma^2)$, if its density is

$$f_W(w) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(w - \mu)^2}{2\sigma^2}\right], \quad -\infty < w < \infty.$$

- ▶ Probabilities are **areas under the density**:
 $\Pr(a \leq W \leq b) = \int_a^b f_W(w) dw.$
- ▶ The curve is symmetric around μ ; σ controls its spread.
- ▶ Standardizing a normal variable gives $(W - \mu)/\sigma \sim N(0, 1)$.

For $Z \sim N(0, 1)$, write its density and cumulative distribution as

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad \Phi(z) = \Pr(Z \leq z) = \int_{-\infty}^z \phi(t) dt.$$

How can we capture an unknown population mean?

In real research, we see **just one** student's sample average, $\bar{Y}$, but **do not** know the true mean μ_Y .

Rather than report only a single estimate, we would like an **interval around it** that usually contains the unknown mean.

The CLT gives us a starting point

For a sufficiently large random sample,

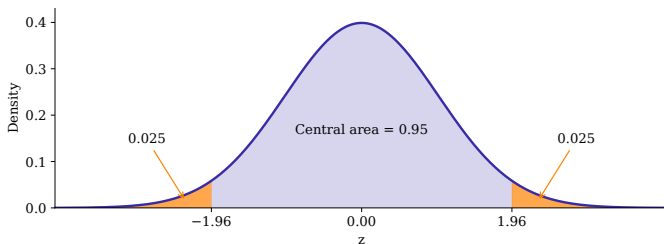
$$\frac{\bar{Y} - \mu_Y}{\text{SD}(Y)/\sqrt{n}} \approx N(0, 1).$$

We can use areas under the standard normal curve to describe how far an estimate is likely to be from its target.

What range contains the middle 95% of the standard normal curve?

That range will help us choose how far to extend our interval on either side of the sample average. We call the resulting range a **confidence interval**.

Why is 1.96 the critical value for a 95% interval?



The 97.5th percentile is $\Phi^{-1}(0.975) = 1.959964 \dots \approx \mathbf{1.96}$. By symmetry, $\Phi(-1.96) \approx 0.025$. Therefore,

$$\Pr(-1.96 \leq Z \leq 1.96) = \Phi(1.96) - \Phi(-1.96) \approx 0.975 - 0.025 = 0.95.$$

A two-sided 95% interval leaves 2.5% in **each** tail.

From the CLT to an interval for the population mean

For large n , the CLT implies

$$\text{Prob}\left(-1.96 \leq \frac{\bar{Y} - \mu_Y}{\text{SD}(Y)/\sqrt{n}} \leq 1.96\right) \approx 0.95.$$

Multiply all three parts of the inequality by $\text{SD}(Y)/\sqrt{n} > 0$:

$$\text{Prob}\left(-1.96 \frac{\text{SD}(Y)}{\sqrt{n}} \leq \bar{Y} - \mu_Y \leq 1.96 \frac{\text{SD}(Y)}{\sqrt{n}}\right) \approx 0.95.$$

Rearrange to put the unknown mean in the middle:

$$\text{Prob}\left(\bar{Y} - 1.96 \frac{\text{SD}(Y)}{\sqrt{n}} \leq \mu_Y \leq \bar{Y} + 1.96 \frac{\text{SD}(Y)}{\sqrt{n}}\right) \approx 0.95.$$

The mean is fixed. The endpoints are random because they depend on the sample.

A standard error estimates the spread of sample averages

The spread of sample averages across random samples is

$$\text{SD}(\bar{Y}) = \frac{\text{SD}(Y)}{\sqrt{n}}.$$

When $\text{SD}(Y)$ is unknown, estimate it with the sample SD, s_Y .

Definition: standard error

The **standard error** is an estimate of the standard deviation of an estimator. For the sample average,

$$\boxed{\text{SE}(\bar{Y}) = \frac{s_Y}{\sqrt{n}} \quad \text{estimates} \quad \text{SD}(\bar{Y}).}$$

With a large random sample, this gives an approximate 95% confidence interval:

$$\boxed{\bar{Y} \pm 1.96 \text{SE}(\bar{Y}).}$$

With small samples, the approximation may be poor and coverage can fall below 95%.

A standard error measures the estimator's sampling noise

Standard deviation: spread of outcomes

s_Y estimates how much *individual district scores* vary around their mean.

Standard error: spread of estimates

$s_Y / \sqrt{n}$ estimates how much *sample means* vary across repeated samples of size n .

For the i.i.d. sample mean,

$$\underbrace{\text{CI length}}_{\text{upper endpoint minus lower endpoint}} = 2 \times 1.96 \times \underbrace{\frac{s_Y}{\sqrt{n}}}_{\text{standard error}} .$$

- ▶ More volatile outcomes give a larger SE and a wider interval, holding n fixed.
- ▶ Quadrupling n roughly halves the SE and interval length, holding dispersion fixed.

The SE is a measure of uncertainty for the estimator over different/repeated samples

Higher confidence requires a wider interval

For confidence level $1 - \alpha$, choose $c = \Phi^{-1}(1 - \alpha/2)$:

$$CI_{1-\alpha} = \bar{Y} \pm c \text{ SE}(\bar{Y}), \quad \text{length} = 2c \text{ SE}(\bar{Y}).$$

Confidence	Each tail	Normal percentile	Critical value c
90%	5%	95th	1.645
95%	2.5%	97.5th	1.960
99%	0.5%	99.5th	$2.576 \approx 2.58$

For the **same sample**, lowering the confidence level produces a narrower interval, at the cost of a greater chance of missing the true mean.

Students construct intervals for the same mean

Return to the university experiment. This time, 3,000 students each report an interval rather than just an average. Try survey budgets of $n = 10, 20, 50, 100$.

1. Each draws n district names using the same bag-and-replacement rule.
2. Researcher j calculates $\bar{Y}_j$, s_j , and

$$CI_j = \left[\bar{Y}_j - 1.96 \frac{s_j}{\sqrt{n}}, \bar{Y}_j + 1.96 \frac{s_j}{\sqrt{n}} \right].$$

3. We check whether CI_j contains the known benchmark $\mu_Y = 654.1565$.

The students use only their own sample to construct their intervals. We use the full population *only to evaluate coverage*.

Before looking at the results: should every student's interval contain the true mean?

One researcher: calculate the mean and standard error

Researcher 1 with $n = 10$ draws these test scores:

638.35, 690.80, 634.10, 649.15, 666.85, 660.10, 679.50, 679.40, 624.55, 664.30

Using the unrounded observations in the dataset,

$$\bar{Y} = \frac{1}{10} \sum_{i=1}^{10} Y_i = 658.710,$$

$$s_Y^2 = \frac{\sum_{i=1}^{10} (Y_i - \bar{Y})^2}{9} = \frac{4267.987}{9}, \quad s_Y = 21.777,$$

$$SE(\bar{Y}) = \frac{21.777}{\sqrt{10}} = 6.886.$$

The sample SD describes the variation among the ten sampled scores. Dividing by $\sqrt{10}$ estimates the sampling SD of their mean.

One researcher: construct and check the interval

Researcher 1's 95% normal-approximation interval is

$$\begin{aligned}\bar{Y} \pm 1.96 \text{SE}(\bar{Y}) &= 658.710 \pm 1.96 \times 6.886 \\ &= 658.710 \pm 13.497 \\ &= [645.213, 672.207].\end{aligned}$$

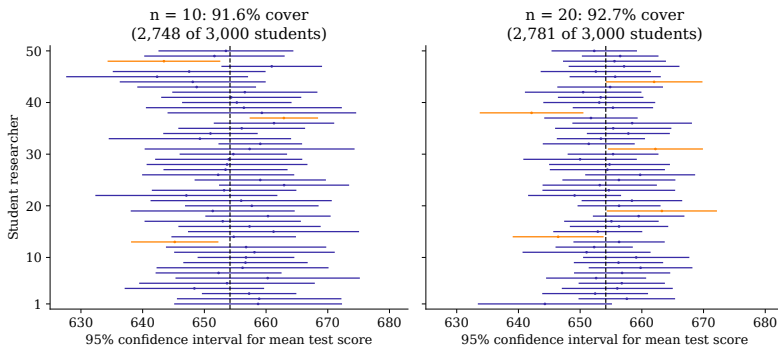
Compare the interval with the true population mean $\mu_Y = 654.1565$.

This realized interval **contains** the true mean.

A different researcher generally gets a different estimate *and* standard error, so both the center and width of the interval can change.

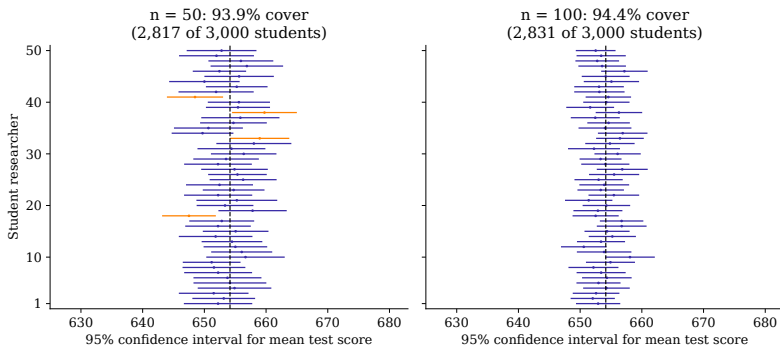
With only ten observations, the normal approximation is imperfect. Calling this a “95% interval” does not guarantee that exactly 95% of such intervals will contain the mean.

Some researchers' intervals miss: $n = 10$ and $n = 20$



Each horizontal segment is one researcher's interval; the dot is the sample mean. **Blue intervals cover** the dashed population mean; **orange intervals miss**. For readability, we plot only the first 50 students. The coverage rate above each panel uses all 3,000 students at that n .

Larger samples give narrower intervals



Again, only the first 50 intervals are plotted; coverage is calculated across all 3,000 students. The horizontal scale is unchanged. Larger samples give narrower intervals, while the overall coverage approaches 95%.

What does “95% confidence” mean?

A statement about the procedure

If we repeatedly draw a random sample and construct an interval in the same way, approximately 95% of those intervals contain the fixed population mean, when the normal approximation is accurate.

- ▶ Before sampling, the interval is random and the population mean is fixed.
- ▶ After sampling, the realized interval either contains the mean or does not. We usually do not know which.
- ▶ It does *not* mean that 95% of individual district scores lie inside the interval.

The experiment makes coverage visible because we know the teaching-population mean. A researcher ordinarily does not have that information.

When should this thought experiment be convincing?

- ▶ **The draws are random.** Every district has the same chance; students do not select the districts they prefer.
- ▶ **The observations are independent.** Replacing and mixing the names makes one draw uninformative about the next.
- ▶ **There is enough information to average out unusual values.** A very skewed population may need a much larger sample before a normal curve works well.

A larger sample helps with sampling noise. It does not fix a survey that systematically leaves out certain kinds of districts.

If students all copied the same district's score n times, would they really have as much information as n independent draws?

From a sample average to an uncertainty statement

1. The **LLN** explains why the sample mean approaches the population mean.
2. The **CLT** gives a normal approximation for the standardized sampling error.
3. The **standard error** estimates the sampling SD of the estimator.
4. A **confidence interval** combines the estimate, SE, and a critical value determined by the confidence level.

$$\text{estimate} \pm \text{critical value} \times \text{standard error}$$

We will apply this same logic to regression coefficients, using standard errors appropriate for those estimators and their assumptions.

Review: sampling uncertainty and confidence intervals

Notion	Main idea or formula	SW section
Sampling distribution	Distribution of $\bar{Y}$ across repeated random samples.	2.5
Standard normal	$N(0, 1)$ has mean 0 and variance 1.	2.4
CLT	$\sqrt{n}(\bar{Y} - \mu_Y) / \text{SD}(Y) \xrightarrow{d} N(0, 1)$ for i.i.d. observations with finite, positive variance.	2.6
Standard error	$\text{SE}(\bar{Y}) = s_Y / \sqrt{n}$ estimates $\text{SD}(\bar{Y})$.	3.2
Confidence interval	$\bar{Y} \pm c \text{SE}(\bar{Y})$; $c = 1.96$ for an approximate 95% interval.	3.3
Coverage	A 95% procedure captures the fixed mean in about 95% of repeated samples when the approximation works well.	3.3

More data: narrower intervals. More variation or higher confidence: wider intervals.

Reference: Stock and Watson, *Introduction to Econometrics*, 4th ed., Chapters 2–3, sections listed above. Textbook contents and section titles.